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HOW DO WE MEASURE TIME?

WE MEASURE TIME BY MEASURING THE CHANGE IN TIME OF AN OBSERVABLE THAT CHANGES IN A PREDICTABLE WAY

THUS, THE UNCERTAINTY Δt WITH WHICH WE MEASURE TIME IS DETERMINED BY THE UNCERTAINTY ΔC WITH WHICH WE CAN MEASURE C

THE UNCERTAINTY Δt IS DETERMINED BY THE RATE WHICH C CHANGE

FOR SUFFICIENTLY SMALL ΔC

$$\Delta C = \left| \frac{dC}{dt} \right| \Delta t$$

WHAT ABOUT A QUANTUM CLOCK?

$$\Delta C = \left| \frac{dC}{dt} \right| \Delta t \Rightarrow \Delta C = \left| \frac{d\langle C \rangle}{dt} \right| \Delta t$$

IF WE ASSUME THAT THE OPERATOR C DOES NOT DEPEND ON TIME

$$\left| \frac{d\langle C \rangle}{dt} \right| = \frac{1}{\hbar} \left| \langle \psi(t) | [C, H] | \psi(t) \rangle \right|$$

$$\frac{d}{dt} \langle A \rangle = \frac{1}{i\hbar} \langle [A, H] \rangle$$

FROM THE UNCERTAINTY RELATION WE ALSO HAVE

$$\Delta C \Delta E \geq \frac{1}{2} \left| \langle \psi(t) | [C, H] | \psi(t) \rangle \right|$$

$$\Delta A \Delta B \geq \frac{1}{2} \left| \langle [A, B] \rangle \right|$$

$$\boxed{\Delta E \Delta t \geq \frac{\hbar}{2}}$$

ENERGY TIME UNCERTAINTY RELATION

WE HAVE SEEN TWO DIFFERENT DESCRIPTIONS OF THE EVOLUTION OF A QUANTUM SYSTEM

$$|\psi(t_2)\rangle = U(t_2, t_1) |\psi(t_1)\rangle$$

SCHRODINGER EQUATION

$$H |\psi(t)\rangle = i\hbar \frac{d}{dt} |\psi(t)\rangle$$

WHAT IS THEIR RELATION

$$H U(t) \cancel{|\psi(0)\rangle} = i\hbar \frac{d}{dt} U(t) \cancel{|\psi(0)\rangle}$$

BASICALLY

$$H U(t) = i\hbar \frac{d}{dt} U(t)$$

← OPERATOR VERSION OF SCHRODINGER EQUATION

$$\omega f(x) = 6 \frac{d f(x)}{dx} \qquad \omega f(x) = 6 \frac{d f(x)}{dx}$$

$$HU(t) = i\hbar \frac{d}{dt} U(t) \Rightarrow \left[\begin{array}{l} U(t) = \exp\left(-\frac{i}{\hbar} H t\right) \\ U(0) = 1 \end{array} \right.$$

$$A = \begin{pmatrix} A_1 & & 0 \\ & \ddots & \\ 0 & & A_d \end{pmatrix} \longrightarrow f(A) \to \begin{pmatrix} f(A_1) & & 0 \\ & \ddots & \\ 0 & & f(A_d) \end{pmatrix}$$

$$f(x) = \omega_0 + \omega_1 A x + \omega_2 x^2 + \dots$$

$$f(A) = \underbrace{\omega_0}_{I\omega_0} + \underbrace{\omega_1 A}_{} + \omega_2 A^2 + \dots$$

$$I\omega_0 = \begin{pmatrix} \omega_0 & & 0 \\ & \ddots & \\ 0 & & \omega_0 \end{pmatrix}$$

$$\omega_1 \begin{pmatrix} A_1 & & \\ & \ddots & \\ & & A_d \end{pmatrix} = \begin{pmatrix} \omega_1 A_1 & & \\ & \ddots & \\ & & \omega_1 A_d \end{pmatrix}$$

$$\omega_2 \begin{pmatrix} A_1 & & \\ & \ddots & \\ & & A_d \end{pmatrix} \begin{pmatrix} A_1 & & \\ & \ddots & \\ & & A_d \end{pmatrix} = \omega_2 \begin{pmatrix} A_1^2 & & \\ & \ddots & \\ & & A_d^2 \end{pmatrix} = \begin{pmatrix} \omega_2 A_1^2 & & \\ & \ddots & \\ & & \omega_2 A_2^2 \end{pmatrix}$$

$$\Rightarrow \omega_N A^N = \begin{pmatrix} \omega_N A_1^2 & & \\ & \ddots & \\ & & \omega_N A_d^2 \end{pmatrix}$$

$$\begin{pmatrix} \underbrace{\omega_0 + \omega_1 A_1 + \omega_2 A_1^2 + \dots}_{f(A_1)} & & \\ & \ddots & \\ & & \underbrace{\omega_0 + \omega_1 A_d + \omega_2 A_2^2 + \dots}_{f(A_d)} \end{pmatrix}$$

$$=$$

$$U(t) = \exp\left(-\frac{i}{\hbar} H t\right)$$

$$H|k\rangle = E_k |k\rangle \qquad U(t) = \left[e^{-i/\hbar \, E_1 t} \right.$$

$[A, f(A)] = 0 \Leftarrow$ given by a power series

$f(A) = \alpha_0 \mathbb{1} + \alpha_1 A + \alpha_2 A^2 + \dots$

Remember $[A, A^n] = 0$

$\hookrightarrow AA^n - A^n A = A^{n+1} - A^{n+1} = 0$

$$[A, f(A)] = [A, \alpha_0 \mathbb{1} + \alpha_1 A + \alpha_2 A^2 + \dots]$$

$$= \alpha_0 [A, \mathbb{1}] + \alpha_1 [A, A] + \alpha_2 [A, A^2] + \dots$$

If B is hermitian $\Rightarrow$ $U = e^{iB}$ is unitary

$\Downarrow$

$$\begin{bmatrix} b_1 & & \\ & \ddots & \\ & & b_d \end{bmatrix} \longrightarrow U = \begin{bmatrix} e^{ib_1} & & 0 \\ & \ddots & \\ 0 & & e^{ib_d} \end{bmatrix}$$

$\forall i \; b_i \in \mathbb{R}$

$|e^{ib_j}| = 1 \quad \forall_j = 1, d$

$\Downarrow$

U unitary

$[A, B] = 0 \Rightarrow \boxed{e^A e^B = e^{A+B}}$ ← not always true

$$\begin{bmatrix} a_1 & & \\ & \ddots & \\ & & a_d \end{bmatrix} \begin{bmatrix} b_1 & & \\ & \ddots & \\ & & b_d \end{bmatrix} \qquad e^A e^B = \begin{bmatrix} e^{a_1} & & \\ & \ddots & \\ & & e^{a_d} \end{bmatrix} \begin{bmatrix} e^{b_1} & & \\ & \ddots & \\ & & e^{b_d} \end{bmatrix} = \begin{bmatrix} e^{a_1 + b_1} & & \\ & \ddots & \\ & & e^{a_d + b_d} \end{bmatrix} = e^{A+B}$$

Rotating a spin

We have seen how to deal with time evolution, but time evolution is only one kind of continuous transformation of states

Consider a spin-1/2 particle in a state $|\psi\rangle$ and imagine rotating

The state is normalized and rotation respects the principle of superposition

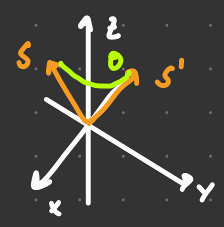
Thus the rotation will be described by a unitary operator

ROTATING SPIN

$R_z(\theta)$

OPERATOR DESCRIBING THE ROTATION OF THE SPIN-1/2 SYSTEM THROUGH AN ANGLE θ ABOUT THE Z-AXIS

WE CONSIDER POSITIVE DIRECTION OF θ TO BE A COUNTER-CLOCKWISE ROTATION WHEN VIEW TOP Z-AXIS



$R_z(\theta)$ OPERATOR DESCRIBING THE ROTATION OF THE SPIN-1/2 SYSTEM THROUGH AN ANGLE θ ABOUT Z-AXIS

$$R_z(\alpha)R_z(\beta) = R_z(\alpha+\beta) \qquad U(t)U(s) = U(t+s)$$

WE START BY OBSERVING THAT $|z\pm\rangle$ SHOULD NOT BE AFFECTED BY THE ROTATION $R_z(\theta)$

$$R_z(\theta)|z+\rangle = e^{i\lambda_+\theta}|z+\rangle \qquad R_z(\theta)|z-\rangle = e^{i\lambda_-\theta}|z-\rangle$$

LAMBDA